Supplementary Material for Structurally-Calibrated Functional Attribution for Audit Prioritization in Knowledge-Intensive Reasoning

Haoran Ma; Ningning Wang

Abstract. This supplementary material records extended proof notes, method details, stage-specific diagnostics, rule-derived audit-category traceability diagnostic, graph-construction ablations, and reproducibility information for the KBS submission package. The LaTeX source bundle remains the authoritative source for mathematical layout; this Word file is a clean readable supplement for portal upload.

Keywords. supplementary material; structural calibration; audit prioritization; rule-derived traceability diagnostic; graph construction; reproducibility

Verification Boundary

MuSiQue KBS-style Knowledge-Audit Details

kbs_style_audit_prioritization_evidence_only

The MuSiQue diagnostic is reported as a constructed-label feasibility demonstration only because audit labels and scoring features are deterministic functions of step type. It is not used as independent audit-prioritization evidence.

The WebQSP diagnostic is reported as a supplementary fixed-schema trace-audit diagnostic. It is not KGQA accuracy evidence, semantic parsing evidence, entity-linking evidence, or production KBS validation.

The rule-derived audit-category traceability diagnostic is zero-human: no human-subjects or human-efficiency claim is made, no deployed KBS workflow is validated, and the evidence level is rule-derived traceability diagnostic. It supports decision traceability and rule-derived category mapping, not human-performance claims.

Supplementary Evidence Map

Table S1. Supplementary evidence map

Diagnostic	Artifact	Status	Allowed interpretation
Synthetic structural stress-test	scripts/run_scu_stress_test.py; outputs/reviewer_v2_experiments	mechanism ablation	QP behavior under designed structural conflict
PRM800K locked split	outputs/real_task_v3_6_prm800k_hash; v3.8 PRM context	positive within boundary	real step-label ranking and PRM800K-like audit prioritization only
Rule-derived traceability diagnostic	outputs/kbs_audit_card_auto_validation_v1	zero-human diagnostic	rule-derived category traceability with circularity limits
Embedding graph-construction ablation	outputs/reviewer_v2_experiments/graph_construction_ablation	graph-interface ablation	TF-IDF is replaceable by semantic embeddings without changing SCU
MuSiQue decomposition diagnostic	outputs/kbs_real_audit_v1	feasibility only	constructed-label pipeline demonstration; no independent audit evidence
WebQSP fixed-schema diagnostic	outputs/webqsp_trace_audit_v1_train/test	diagnostic only	separability and metric-artifact diagnostic, not KGQA performance
Countries-KG typed-edge stage	outputs/kbs_ontology_edge_pilot/kg_pilot_report.json	typed-edge diagnostic	typed ontology edges can change structural diagnostics; no deployed workflow validation

Rule-Derived Audit-Category Traceability Diagnostic

Failure-mode localization is evaluated as an information-retrieval task over traces. The scalar condition uses only the w_struct score, while the SC-FMA condition uses fidelity, structural necessity, redundancy, bottleneck status, and recommended action fields. Targets are rule-derived from PRM800K labels, ranking disagreement, redundancy density, bottleneck flags, and the existing failure taxonomy; they are not human adjudications.

Table S2. Rule-derived audit-category traceability diagnostic

View	Recall@25 %	NDCG@25 %	MRR	Top-1 hit	Cost proxy

w_struct scalar only	0.235	0.353	0.354	0.000	0.0007
SC-FMA decomposition	0.699	0.978	1.000	1.000	0.0002

Interpretation: the decomposition fields recover rule-defined audit targets more effectively than the scalar-only view. This is decision-traceability and information-recovery evidence, not human-subject or human-efficiency evidence.

Graph Construction and Embedding Ablation

The graph-construction ablation compares temporal-only, TF-IDF topical, Jaccard topical, Sentence-BERT embedding topical, and shuffled-topical variants under the same diagnostic protocol. The embedding diagnostic uses sentence-transformers/all-MiniLM-L6-v2 through sentence-transformers version 3.4.1 with 384-dimensional vectors. The report records the HuggingFace cache path, download/cache status, runtime status, and graph-constructor metadata.

Table S3. Graph construction and embedding ablation

Constructor	Spearman rho	Kendall tau	Faithfulness	Edges
embedding_topical	0.0615	0.0541	0.0863	2,282
temporal_only	0.0596	0.0525	0.0507	2,098
jaccard_topical	0.0517	0.0459	0.0557	1,655
tfidf_topical	0.0515	0.0458	0.0542	1,648
shuffled_topical	0.0515	0.0458	0.0542	1,648

Interpretation: the result is graph-interface robustness evidence. TF-IDF is a lightweight default constructor; Sentence-BERT embeddings can be substituted at the graph interface. The manuscript does not claim a production graph-ranking or live KBS performance result.

Hyperparameter and Stress-Test Diagnostics

The QP hyperparameter sensitivity grid sweeps gamma in {0.0, 0.1, 0.2, 0.5, 0.8} and delta in {0.0, 0.05, 0.1, 0.2, 0.4}. The diagnostic default is gamma=0.2 and delta=0.1. The grid is supplementary QP robustness evidence only; it is not positive external validation and does not change the PRM800K Ridge recommendation.

Table S4. Alpha/beta QP sensitivity grid

alpha	beta=0.0	beta=0.5	beta=1.0
0.5	0.6393 (+0.0462)	0.5714 (-0.0217)	0.5097 (-0.0834)
1.0	0.5605 (-0.0326)	0.5931 (+0.0000)	0.5585 (-0.0345)
2.0	0.4868 (-0.1063)	0.5346 (-0.0585)	0.5572 (-0.0359)

Finding Interpretation

Removing bottleneck pressure lowers Spearman relative to the default on the stress-test generator. The bottleneck term is useful in the synthetic regime where labels encode bottleneck protection.

Moderate-to-strong redundancy and bottleneck weights remain competitive on the structural stress-test. The stress-test behavior is not a single-point hyperparameter artifact.

PRM800K still favors w_struct and the Ridge approximation. QP is reserved for high-structure-conflict settings; Ridge is recommended when learned fidelity is already strong.

PRM800K Variant and Audit Readout

The strongest real-data signal is w_struct. SC-FMA Ridge is a preservation-with-decomposition approximation: it closely preserves w_struct while exposing fidelity, structural necessity, redundancy, and bottleneck fields for audit cards. QP remains useful for controlled structural-conflict settings but is downgraded on PRM800K.

Table S5. PRM800K variant comparison

Method	Mean Spearman	Audit interpretation
w_struct	0.611	Primary PRM800K step-ranking signal
SC-FMA Ridge	0.604	Closest SC-FMA preservation-with-decomposition approximation
SC-FMA QP	0.438	Downgraded on PRM800K; reserved for structural-conflict settings
Raw local utility	0.399	Weaker direct ranking baseline
Random	0.000	Control

WebQSP Fixed-Schema Trace-Audit Diagnostic

Development used 494 train traces and 2,964 steps. The locked diagnostic used 1,627 test traces and 9,762 steps. Data-audit counts for missing entities, disconnected subgraphs, duplicate traces, and invalid logical forms are recorded before metric computation.

Table S6. WebQSP fixed-schema trace-audit diagnostic

Method	Spearman rho	Pairwise accuracy	NDCG@25%
Raw rule delta	1.000	1.000	1.000
SC-FMA Ridge	0.878	1.000	1.000
Relative position	0.878	0.667	1.000
SC-FMA QP	0.293	0.556	0.613
Graph degree	-0.878	0.111	0.387
Random	0.000	0.500	0.500

Safe interpretation: the fixed six-step schema makes replay-derived importance largely separable by step role and position, making complex policy optimization redundant under this schema. This is a transferability and evaluation-confound diagnostic, not evidence that SC-FMA improves KGQA, semantic parsing, or live KBS maintenance.

MuSiQue KBS-Style Knowledge-Audit Details

The MuSiQue diagnostic constructs trace records from decomposition evidence chains. The locked split contains 1,415 samples and 12,715 steps. No API calls are used. The field-level leakage check marks the diagnostic as feasibility-only because labels and scoring features share deterministic step-type sources.

Table S7. MuSiQue KBS-style knowledge-audit feasibility demonstration

Method	Spearman rho	Top-1 hit	NDCG@25%	Boundary
w_struct	0.9652	1.0000	1.0000	feasibility only
SC-FMA Ridge	0.9653	1.0000	1.0000	feasibility only
Raw CIU	0.9004	1.0000	1.0000	feasibility only
Random	0.0000	0.0000	0.5000	control

Reproducibility Notes

- All reported diagnostics are offline and use zero API calls unless explicitly documented otherwise.
- The controlled synthetic stage is reproducible from scripts/generate_synthetic_traces.py with fixed seeds.
- The structural stress-test is reproducible from scripts/run_scu_stress_test.py and scripts/run_scu_hyperparameter_sensitivity.py.

QP hyperparameter sensitivity is reproduced with scripts/run_scu_hyperparameter_sensitivity.py. Outputs are archived as outputs/reviewer_v2_experiments/scu_hyperparameter_sensitivity/scu_hyperparameter_sensitivity.json, outputs/reviewer_v2_experiments/scu_hyperparameter_sensitivity/scu_hyperparameter_sensitivity.md, outputs/reviewer_v2_experiments/scu_hyperparameter_sensitivity/alpha_beta_grid.csv, and outputs/reviewer_v2_experiments/scu_hyperparameter_sensitivity/gamma_delta_grid.csv. The alpha/beta grid fixes gamma=0.2 and delta=0.1 and sweeps alpha in {0.5, 1.0, 2.0} by beta in {0.0, 0.5, 1.0}; it is a supplementary QP diagnostic only, not positive external validation.

- Rule-derived audit-category traceability diagnostic is reproducible from scripts/run_audit_card_auto_validation.py and outputs/kbs_audit_card_auto_validation_v1.
- Embedding graph-construction ablation is reproducible from scripts/run_graph_construction_ablation.py and outputs/reviewer_v2_experiments/graph_construction_ablation.
- The WebQSP diagnostic is reproducible from scripts/run_webqsp_trace_audit.py, scripts/validate_webqsp_trace_audit.py, and scripts/analyze_webqsp_separability.py.
- The Countries-KG typed-edge stage is reproducible from src/fma/graph/kg_ontology_pilot.py and outputs/kbs_ontology_edge_pilot/kg_pilot_report.json.

Claim Boundary Summary

This supplementary file supports the manuscript title: Structurally-Calibrated Functional Attribution for Audit Prioritization in Knowledge-Intensive Reasoning

Allowed: bounded KBS-facing audit-prioritization methodology; PRM800K-like step-label ranking and audit context; zero-human rule-derived traceability diagnostic for decomposition information gain; graph-interface ablation with Sentence-BERT embeddings; diagnostic WebQSP and Countries-KG typed-edge stages; MuSiQue feasibility demonstration.

Not claimed: downstream PRM/filtering gains, GSM8K/HotpotQA replay-pass evidence, WebQSP KGQA improvement, production knowledge-base deployment validation, MuSiQue-derived independent proof, human-subject evidence, or human-performance improvement.

Revision-Specific Boundary Notes

- SCU component contribution on a structural stress-test benchmark: the stress-test labels explicitly encode redundancy-awareness and bottleneck-protection, so QP gains are interpreted as intended behavior in a designed regime, not general ranking superiority.
- Countries-KG Typed-Edge Stage: the Countries-KG typed-edge stage is a typed-edge construction stage only; it does not validate live KBS deployment, KGQA accuracy, semantic parsing, or maintenance outcomes.
- Rule-derived audit-category traceability diagnostic: targets are derived from PRM800K labels, ranking disagreement, redundancy density, bottleneck flags, and failure-taxonomy rules. Weak utility-anchor targets have no or low circularity; redundancy consolidation, bottleneck protection, and structural over-correction have partial circularity and are traceability diagnostics only.
- The w_struct regressor is a frozen 15-feature Ridge model; the LaTeX source lists the exact feature names and the verifier checks that list against the frozen model JSON.